



PCR Kit

Catalog No. PG-30013**Quantity:** 200 reactions**Reaction volume:** 25 ul/reaction

Application

Taq DNA polymerase is used combination with PCR buffer provides robust performance for reproducible results in a wide range of PCR applications without the need for time-consuming optimisation. The PCR kit includes a ready-to-use premixed solution containing Taq DNA polymerase, PCR buffer, and dNTPs.

Kit Components

1. Taq PCR 2X Premix
2. ddH₂O

Storage

Store at -20°C.

Related Products

Taq DNA polymerase	PG-30001
Hotstart Taq DNA polymerase	PG-30002
Pr Taq DNA polymerase (Taq+Pfu)	PG-30003
Hotstart Pr Taq DNA polymerase (Taq+Pfu)	PG-30004
Klen Taq DNA polymerase	PG-30005
Hotstart Klen Taq DNA polymerase	PG-30006
Anti-Taq Hotstart Antibody	PG-30007
Real-time PCR kit	PG-30009
One Step RT-PCR kit	PG-30010
Genotyping kit	PG-30012

Table 1. Reaction composition using PCR kit

Component	Volume/reaction	Final concentration
Taq PCR 2X Premix:	25 µl	1 x PCR buffer 200 µM of each dNTPs
Forward primer	Variable	0.1-0.5 µM
Reverse primer	Variable	0.1-0.5 µM
Template RNA	Variable	-
Total volume	50 µl	

Table 2. Thermal cycler conditions:

			Additional comments
Initial PCR activation step:	3min	94°C	
3-step cycling:			
Denaturation:	0.5-1 min	94°C	
Annealing:	0.5-1 min	48-68°C	Approximately 5 °C below T _m of primers
Extension:	1min/per Kb DNA	72°C	Use an extension time of approximately 1 min per Kb DNA
Number of cycles:	30-40		
Final extension	3-10 min	72°C	

Table 3. Guidelines for the design primers:

Sequence: ★ Avoid runs of 3 or more G or C at 3'end
★ Avoid a T at 3'end
★ Avoid mismatched at the 3' end
★ Avoid complementary sequences with a primer and between primers

Length: 18-30 nucleotides
GC content: 40-60%
T_m: $T_m = 2^{\circ}\text{C} \times (A+T) + 4^{\circ}\text{C} \times (C+G)$
Conc.: 0.1-0.5 µM (0.2 µM)